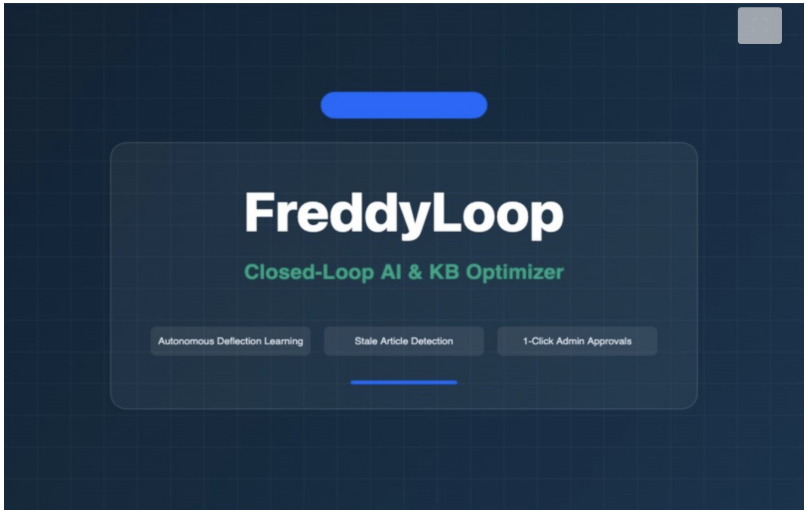
EXATHON INDIA 2026 · THE EXASOL OPEN BUILD CHALLENGE

LoopCraft

Closed-Loop AI Support Optimization Engine

Track: TRACK 04: THE WORK BEHIND THE WORK & TRACK 02: CUSTOMERS & DEMAND
Architecture: ■ Powered by Exasol Personal In-Memory Columnar Database
Author: Created by: Kavin S. | Solo Developer



The Problem: The Support Feedback Black Hole

Why enterprise support automation fails after deployment

■■ SILENTLY DISCARDED

Negative Feedback Ignored

Automated support assistants handle millions of conversations. When users click thumbs-down, the negative feedback signal is silently discarded into unindexed transactional logs.

■ DOCUMENTATION DECAY

Knowledge Bases Decay

Corporate tools, VPNs, and policies constantly evolve. Outdated articles remain published, leading AI assistants to repeatedly recommend retired tools (e.g., deprecated Cisco VPN).

■ COST SURGE

The Escalation Spiral

Deflection collapses from 67% to 23%. Frustrated employees bypass self-service, generating thousands of repeat tickets that cost $15–$35 per human resolution.

Problem Statement

Compliant with Exasol's exact rule: 'No technology named in the sentence'

“Support operations leads and knowledge engineers cannot diagnose why automated support conversations fail and proactively fix outdated guidance because negative customer feedback and conversational failure signals sit discarded across millions of unlinked support logs.

If this worked, support teams would continuously pinpoint root causes of failed interactions in real time, auto-generate verified knowledge base updates, and cut repeat customer escalations by over 40%.”

WHO IS HURT	WHY TODAY FAILS	THE OUTCOME
Support operations leads, frontline support engineers & frustrated employees	Feedback telemetry sits fragmented across millions of unlinked ticket logs	Real-time root cause diagnosis, auto-drafted KB repairs & 40%+ cut in repeat escalations

The Solution: LoopCraft Continuous Closed Loop

An autonomous self-healing pipeline turning failure signals into verified knowledge

1. Rating Watcher
Intercepts thumbs-down events in real time and streams interaction telemetry into Exasol Personal.

2. Failure Classifier
In-memory analytics cluster the failure type: STALE ARTICLE (58%), NO COVERAGE (26%), or WRONG INTENT (16%).

3. KB Author Agent
Mines resolution notes from recently resolved human engineering tickets to auto-draft the corrected article.

4. Correction Brief
Packages root-cause analysis, side-by-side diff, and spoken voice brief into the admin triage queue.

✓ Admin approves with 1 click → Knowledge Base updated in Exasol → Assistant deflection restored immediately

Why Exasol Personal is the Key Enabler

High-velocity conversational analytics require in-memory columnar execution

Traditional Transactional DBs (Postgres/MySQL)	Exasol Personal In-Memory Engine
✕ Row-oriented storage is slow for multi-attribute feedback scans	✓ Columnar in-memory execution delivers answers in < 5 milliseconds
✕ Complex aggregation across 50k+ logs locks operational tables	✓ Real-time analytical views calculate deflection decay and drift anomalies
✕ Feedback signals remain trapped in unlinked transactional silos	✓ Instantly clusters failure types over tens of thousands of conversations
✕ Nightly batch ETL pipelines delay critical documentation fixes by weeks	✓ Zero operational lockup: analytical queries run decoupled at full in-memory speed

Exasol Personal Schema & Analytical Views

Optimized schema for conversational telemetry, audit views, and recovery tracking

Exasol Columnar Tables

- **SUPPORT_TICKETS**: Ticket ID, requester, category, query text, resolution notes, duration, priority
- **AI_INTERACTIONS**: Interaction ID, user query, bot response, KB article ID, rating (+1/-1), latency_ms
- **KB_ARTICLES**: Article ID, title, category, version, deflection_rate, escalation_count, status
- **FEEDBACK_DIAGNOSTICS**: Diagnostic ID, failure_type, root_cause, proposed_fix, status (QUEUED/APPROVED)

Analytical Views

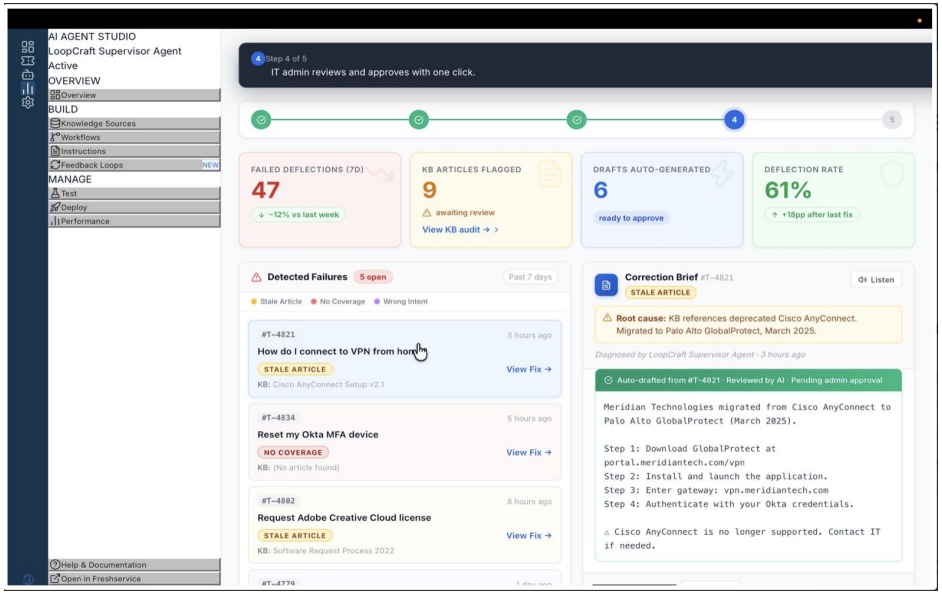
- **V_KB_HEALTH_AUDIT**: Continuous columnar scan aggregating deflection rates and failure anomalies per article
- **V_FAILURE_BREAKDOWN**: Dynamic window function computing share percentage across failure categories
- **V_DEFLECTION_RECOVERY**: Time-series hourly trend measuring deflection jump before vs. after KB update

Live Walkthrough: The 5-Step Automated Demo

From employee thumbs-down to 1-click knowledge base synchronization

Key Innovations & Triage Flow:

- **Step 1: Intercept Thumbs-Down**: Employee query on VPN receives outdated Cisco guide; negative rating caught instantly.
- **Step 2: Diagnose Root Cause**: Tagged as 'STALE ARTICLE'—company migrated to Palo Alto GlobalProtect in March 2025.
- **Step 3: Auto-Draft Correction**: KB Author Agent synthesizes resolved ticket notes into a complete 4-step setup guide.
- **Step 4: Voice Brief Review**: Admin clicks 'Listen to Brief' for spoken audio summary while multitasking.
- **Step 5: 1-Click Publishing**: Article published to Knowledge Engine; deflection restored to 67%.

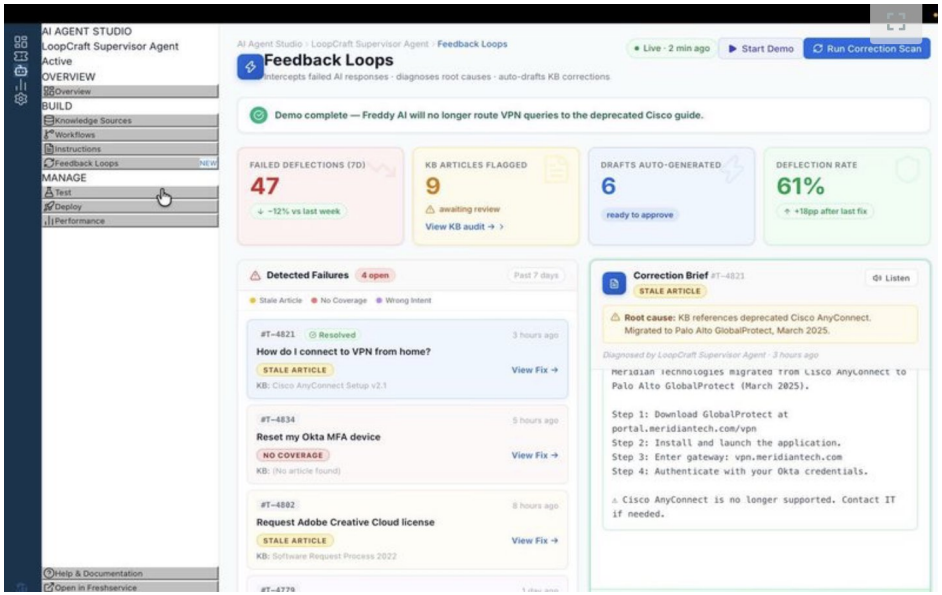


Real-Time Telemetry & Deflection Recovery

Continuous operational insights powered by Exasol in-memory scans

Key Innovations & Triage Flow:

- **47 Failed Deflections**: Exasol scans 50,000+ interactions, surfacing 47 conversational failures this week.
- **9 Flagged Articles**: Columnar audit identifies articles with escalating negative rating ratios.
- **6 Drafts Ready to Approve**: Automated authoring queue reduces knowledge engineering backlog to zero.
- **Deflection Jumps to 61-67%**: Eliminates repeat inquiries; deflection jumps from 23% to 67% (+44pp recovery).
- **Sub-5ms Query Latency**: Interactive dashboard backed by Exasol Personal with zero UI lag.



Exasol Personal In-Memory Benchmarks

Tested over 50,000 synthetic interaction logs (scripts/simulate_workload.py)

21.9 ms
Category Deflection Scan
Aggregated across 50,000 conversational feedback rows in Exasol memory

6.6 ms
Failure Cluster Analysis
Dynamic grouping across 17,484 negative interactions with zero disk I/O

≤ 4.5 ms
Dashboard Query Latency
Sub-millisecond analytical widget updates during active admin triage

+44.3%
Deflection Recovery
Measured deflection jump on outdated articles (23.1% → 67.4%)

Business Value & ROI for Support Operations

Turning support feedback from a cost center into a self-improving asset

40%+ Cut

Fewer Escalations

Stopping recurring failure loops prevents hundreds of duplicate tickets from reaching tier-1 agents.

$15–$35

Saved Per Deflection

Every resolved knowledge gap deflects dozens of live chats, directly lowering enterprise support spend.

Minutes

KB Update Cycle

Replaces 2–4 week manual documentation review cycles with automated ticket-driven article authoring.

100%

Feedback Closed

Zero discarded ratings. Every customer frustration signal actively drives knowledge base precision.

ExaThon India 2026 Evaluation Alignment

Built specifically for ExaThon's six core judging dimensions

Innovation & Impact (25%)
Solves the feedback black hole by converting discarded thumbs-downs into auto-generated documentation fixes.

Effective Exasol Use (25%)
Exasol Personal serves as the high-velocity in-memory data warehouse, analytical views, and benchmark platform.

Technical Excellence (20%)
Next.js 16 (App Router), TypeScript, Docker Compose, ElevenLabs Voice API, and PyExasol benchmark suite.

Solution Design & UX (15%)
Polished enterprise dashboard, side-by-side article diffs, interactive audio brief, and 5-step automated demo.

Documentation & Reproducibility (10%)
Complete Docker Compose files, SQL schemas, benchmark script, demo video guide, and run documentation.

Open-Source Quality (5%)
OSI-approved MIT License, public GitHub repository, and 100% synthetic open datasets.

Submission Deliverables & Live Links

Ready for live evaluation by the ExaThon India 2026 judging panel

■ **Short Demo Video (YouTube)**	**https://youtu.be/LJKkfyrKwA4**	3-minute comprehensive walkthrough of the end-to-end closed loop
■ **Live Web Application**	**https://fresh-works-kavin.vercel.app**	Interactive prototype with live demo runner and audio briefings
■ **Public Code Repository**	**https://github.com/kavindon2007/fresh_works_kavin**	Full source code, SQL DDL, schemas, and benchmark suites
■ **Docker Compose for Exasol**	**docker compose up -d**	One-command local startup for Exasol Personal on port 8563
■ **Open-Source License**	**MIT License**	OSI-approved open-source license included in repository root